

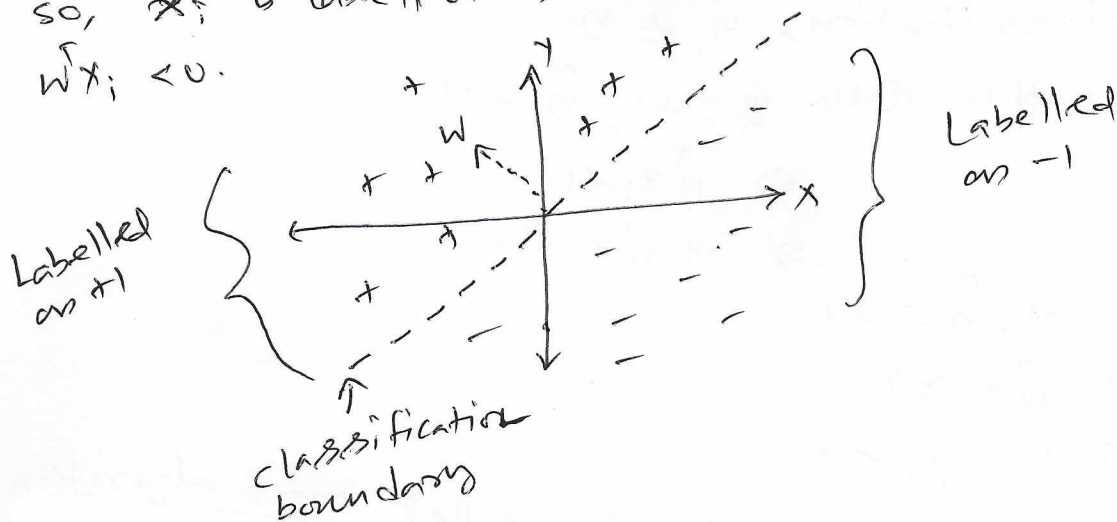
Perceptron

Linear deterministic binary classification method. Linear function of the form $w^T x$ to perform binary classification. class labels: $+1, -1$.

Prediction: ~~$y_i = \text{sgn}(w^T x_i)$~~ $y_i = \text{sgn}(w^T x_i)$

$\text{sgn}(x)$: signum function, returns 1 if x is ≥ 0 , and -1 if $-ve$.

so, x_i is labelled as $+1$ if $w^T x_i \geq 0$, and labelled -1 if $w^T x_i < 0$.

Finding parameter w :

1. Start with a random w_0 (let's consider $w_0 = 0$)
2. Randomly select a data point (x_i, y_i)
3. If the predicted label for the point is wrong, update w as the algorithm below to correct the misclassification. Note: data point x_i is misclassified if $y_i \cdot w^T x_i < 0$.
4. Repeat steps 2 & 3 until there are no updates in w for a continuous fixed number of iterations or the number of iterations crosses a predicted threshold.

Algo: finding w

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 $w_0 = 0$ 
for  $t = 0 \dots \text{maxIter}$  do
     $i = \text{rand}(1, \dots, n)$ 
    if  $y_i \cdot w^T x_i < 0$ 
         $w_{t+1} \leftarrow w_t + y_i \cdot x_i$ 
    else
         $w_{t+1} \leftarrow w_t$ 
end
end

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Why $y_i w^T x_i < 0$?

① For correct classification, $\hat{y}_i = y_i$. Now, $\hat{y}_i = \text{sgn}(w^T x_i)$

↓

$$\text{If } y_i = +1; w^T x_i \geq 0$$

$$\text{If } y_i = -1; w^T x_i < 0$$

So, if $y_i = +1$ & $w^T x_i \geq 0$, then $y_i w^T x_i \geq 0$

If $y_i = -1$ & $w^T x_i < 0$, then $y_i w^T x_i \geq 0$

② for misclassification, $\hat{y}_i \neq y_i$

This can happen when $y_i = +1, \hat{y}_i = -1$

$$\Rightarrow w^T x_i < 0$$

$$\Rightarrow y_i w^T x_i < 0$$

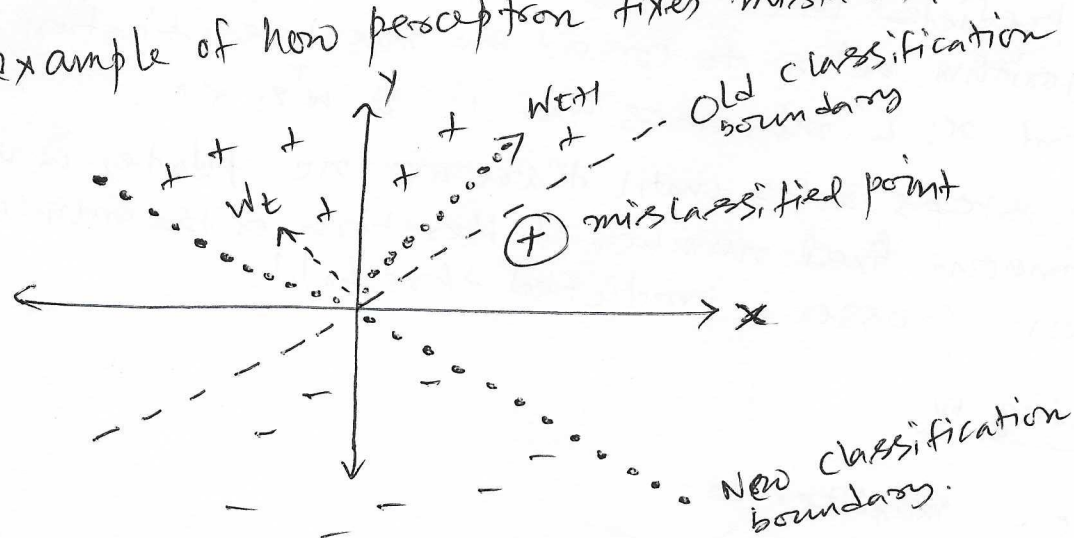
or, $y_i = -1, \hat{y}_i = +1$

$$\Rightarrow w^T x_i \geq 0$$

$$\Rightarrow y_i w^T x_i \leq 0$$

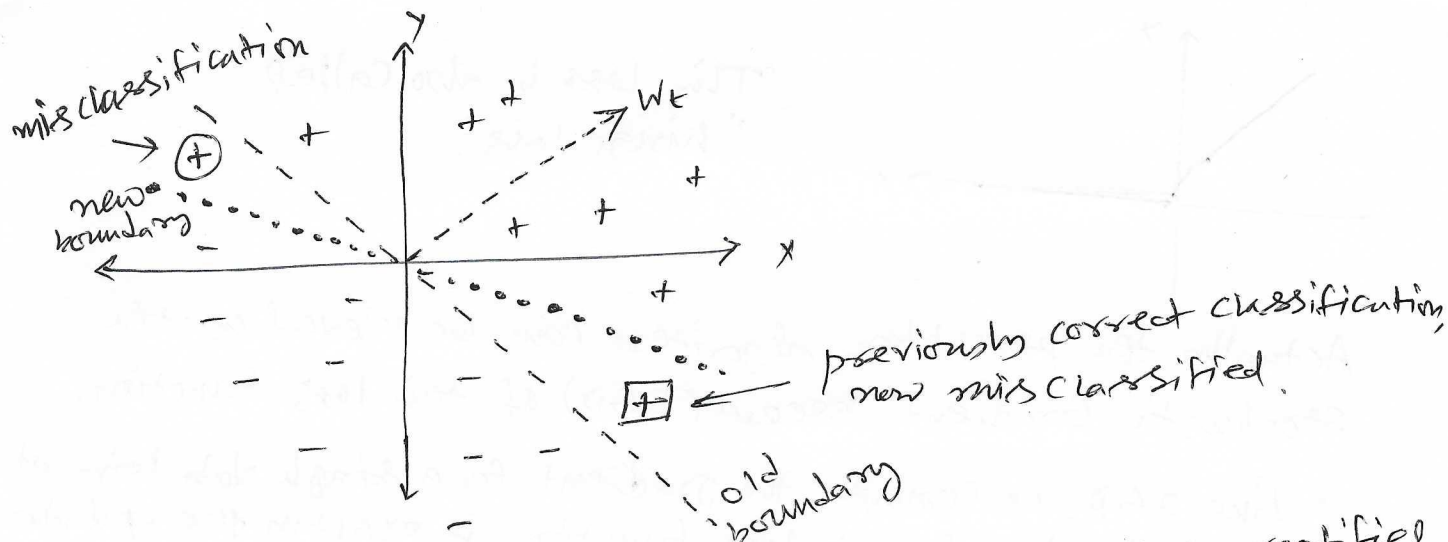
Note: Linear & logistic regression are called batch algorithms as they require the whole dataset to be processed together. Perceptron algorithm is an online algorithm which uses the data points one-by-one.

One example of how perceptron fixes mistakes:



So the perceptron model corrected the classification boundary so that w_{new} correctly predicts the previously misclassified point. But this might not always be the case.

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In this case, although the misclassification has been rectified, one point which was correctly classified before, is now misclassified now.

↓ Problem.

The dataset is not linearly separable.

Perceptron is a linear classifier, and can't handle this.

It can be shown that the perceptron algorithm improves the value of $y_i w_{t+1}^T x_i$, making it more positive with every update:

$$w_{t+1} = w_t + y_i x_i$$

$$\begin{aligned} \text{Now, } y_i w_{t+1}^T x_i &= y_i ((w_t + y_i x_i)^T x_i) \\ &= y_i w_t^T x_i + y_i^2 x_i^T x_i \\ &= y_i w_t^T x_i + \|x_i\|^2 \end{aligned}$$

Since $\|x_i\|^2 > 0$, $y_i w_{t+1}^T x_i > y_i w_t^T x_i$. Hence the model is improved.

Loss function

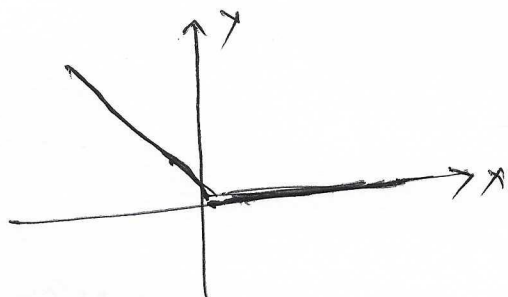
Our goal is to find w that maximizes $y_i w^T x_i$. This can be written as,

$$w^* = \arg \min_w \sum_{i \text{ misclassified}} -y_i w^T x_i$$

The loss is, $L_{\text{perceptron}}(w; x_i, y_i) = \begin{cases} -y_i w^T x_i & \text{if } y_i w^T x_i < 0 \\ 0 & \text{otherwise} \end{cases}$

So we can write this as,

$$L_i = \max \{0, -y_i w^T x_i\}, \text{ \& } L = \sum_i L_i$$



This loss is also called "hinge loss".

Actually the perceptron algorithm can be viewed as the stochastic Gradient Descent (SGD) of this loss function.

- Like SGD, we compute the gradient for a single data point at a time.
- The derivative of this loss function is exactly the update we have been using so far.

↓

$$w_{t+1} = w_t - \nabla_w L_i$$

$$\nabla_w L_i = \begin{cases} 0, & \text{if the prediction is correct} \\ -y_i x_i, & \text{if the prediction is incorrect.} \end{cases}$$

(Since gradient of L is $y_i x_i$, this is the direction of steepest decrease/increase. That's why w_{t+1} is updated with $y_i x_i$, i.e. $w_{t+1} = w_t + y_i x_i$)

Proof of convergence of perceptron.

Assume $\|x_i\| \leq 1 \ \forall i$, and the data is linearly separable, i.e. there exists a unit vector w^* with margin of separation γ , then the perceptron algorithm always finds a correct solution in at most $1/\gamma^2$ mistakes.

$\Rightarrow \exists$ some vector w^* that classifies everything correctly means,
 $y_i w^* x_i > 0 \ \forall i \rightarrow \textcircled{1}$

Margin measures how far the closest point is from the separating line. Here the distance of the closest point to the line w^* is γ .
 large $\gamma \rightarrow$ classes are well separated, small $\gamma \rightarrow$ classes almost touch.

so if we normalize w^* so that $\|w^*\| = 1$, then,

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$$\gamma = \min_i y_i w_i^* x_i$$

$$\Rightarrow y_i w_i^* x_i \geq \gamma \quad \forall i \rightarrow (2)$$

Now let's back to the proof.

Perceptron update rule: $w_{t+1} = w_t + y_i x_i$

Let M be the total # of mistakes. Assume we start with $w_0 = 0$.
We have to prove $M \leq 1/\gamma^2$

The proof has 2 key inequalities:

Step 1: Lower bound on alignment with w^*

We measure how aligned w_t becomes with the true separator w^* .

$$\begin{aligned} w_{t+1}^T w^* &= (w_t + y_i x_i)^T w^* \\ &= w_t^T w^* + y_i x_i^T w^* \end{aligned}$$

$$\text{Using (2), } w_{t+1}^T w^* \geq w_t^T w^* + \gamma \rightarrow (3)$$

So every mistake increases alignment by at least γ . Now apply this repeatedly:

$$\text{Start from } w_0 = 0 \Rightarrow w_0^T w^* = 0$$

$$\text{Mistake \#1} \Rightarrow w_1^T w^* \geq w_0^T w^* + \gamma \geq \gamma$$

$$\text{Mistake \#2} \Rightarrow w_2^T w^* \geq w_1^T w^* + \gamma \geq \gamma + \gamma \geq 2\gamma$$

$$\vdots$$

$$\text{Mistake \#M} \Rightarrow w_M^T w^* \geq M\gamma \rightarrow (4)$$

So every mistake pushes w at least γ units towards correct direction. After M mistakes, we have moved at least $M\gamma$ towards w^* .

Step 2: upper bound on norm growth

We now bound how large $\|w_M\|$ can become.

$$\|w_{t+1}\|^2 = \|w_t + y_i x_i\|^2 = \|w_t\|^2 + 2y_i w_t^T x_i + \|x_i\|^2$$

Update happens only on mistakes. So,

$$y_i w_t^T x_i \leq 0 \Rightarrow 2y_i w_t^T x_i \leq 0$$

$$\text{So, } \|w_{t+1}\|^2 \leq \|w_t\|^2 + \|x_i\|^2 = \|w_t\|^2 + 1 \quad (\text{since } \|x_i\| \leq 1)$$

$\rightarrow (5)$

Now apply repeatedly:

$$\begin{aligned} \text{Start from } w_0 = 0 &\Rightarrow \|w_0\|^2 = 0 \\ \text{Mistake \#1} &\Rightarrow \|w_1\|^2 \leq \|w_0\|^2 + 1 \leq 1 \\ \text{Mistake \#2} &\Rightarrow \|w_2\|^2 \leq \|w_1\|^2 + 1 \leq 2 \\ &\vdots \\ \text{Mistake \#M} &\Rightarrow \|w_M\|^2 \leq M \rightarrow \textcircled{6} \end{aligned}$$

Step 3: Combine both results

$$\text{From step \#1 eq. \textcircled{4}: } w_M^T w^* \geq M\gamma$$

We use the following inequality (a.k.a. Cauchy Schwarz inequality)

$$a^T b \leq \|a\| \cdot \|b\|$$

$$\text{So, } w_M^T w^* \leq \|w_M\| \cdot \|w^*\| \leq \|w_M\| \quad (\text{Since } \|w^*\| = 1)$$

$$\text{Using \textcircled{4}, } M\gamma \leq \|w_M\|$$

$$\text{Using norm bound from \textcircled{6}, } M\gamma \leq \sqrt{M}$$

$$\Rightarrow M^2 \gamma^2 \leq M \Rightarrow M \gamma^2 \leq 1$$

$$\Rightarrow \boxed{M \leq 1/\gamma^2}$$

So, every mistake pushes w_t towards w^* . But the norm of w_t grows only like $\sqrt{M}$.